

		Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Department: AI-ML/AI-RO/AI/AI-DS/CSE/MICRO/SAP/QUICK/ORACLE/IT/AERO		
Subject Name	PROBABILITY, STATISTICS AND NUMERICAL METHODS	A.Y	2025/2026	
Subject Code	303191251	Semester	4th	
Tutorial-2				
Sr No	Question	COs	B.T	Competence
1	A box contains 6 red, 4 blue, and 5 green balls. If one ball is drawn at random, find the probability that it is (a) red, (b) blue, (c) not green.	2	2	Understand
2	In a class of 50 students, 30 like Maths, 28 like Physics, and 18 like both. Find the probability that a randomly selected student likes (a) Maths only, (b) Physics only, (c) at least one subject.	2	3	Apply
3	A card is drawn from a deck of 52 cards. Find the conditional probability that it is a king given that it is a face card.	2	3	Apply
4	A machine produces 5% defective items. If 3 items are selected at random, find the probability distribution of number of defective items.	2	4	Analyze
5	A bag contains 3 white and 2 black balls. Two balls are drawn one after the other without replacement. Use Bayes’ theorem to find the probability that the first drawn ball was white if the second drawn ball is white.	2	4	Analyze
6	A random variable X has the following probability distribution: X: 0,1,2,3 ; P(X): 0.1, 0.3, 0.4, 0.2. Find (a) E(X), (b) Var(X).	2	3	Apply
7	The mean of a binomial distribution is 6 and variance is 2.4. Find n and p.	2	5	Evaluate
8	A Poisson distribution has mean 4. Find P(X=0), P(X≤2), and P(X≥3).	2	3	Apply
9	Heights of students follow normal distribution with mean 165 cm and SD 10 cm. Find the probability that a student chosen at random has height between 160 cm and 180 cm.	2	4	Analyze
10	If a continuous random variable X has PDF f(x)=k x ² on (0,2), find (a) value of k, (b) mean of X, (c) variance of X.	2	6	Create